Bucket Brigade: A Parametric Cooperative-Competitive Benchmark with Computable Nash Equilibrium Structure

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June 11, 2026

Abstract

Canonical cooperative-MARL benchmarks (Overcooked, Hanabi, SMAC, Melting Pot, MAgent, PettingZoo) trade equilibrium transparency for environment richness: their state spaces are large enough that no Nash equilibrium is computable, so when a new algorithm beats PPO on Overcooked the reader cannot tell whether the algorithm has converged to the "right" joint policy or merely to a Pareto-different one. *Bucket Brigade* inverts this trade. It is a finite parametric cooperative-competitive stochastic game on a ring of houses whose per-agent action space at the minimal parameterization is 8, whose state space is 2304 states, and whose three load-bearing scalars (β, κ, c) control equilibrium structure continuously across cells. We give the environment specification, derive closed-form boundary inequalities for the three Nash-equilibrium regimes (symmetric all-Work, asymmetric 1-Worker, no-pure-NE collapse) via a single-house mean-field reduction, and report empirical agreement between predicted and PPO-realised verdicts on a 39-cell phase diagram $(\beta \times \kappa \times c)$. The reduction correctly predicts the qualitative phase order (collapse $\rightarrow$ symmetric $\rightarrow$ mixed $\rightarrow$ asymmetric) and β-independence; the 39-cell grid surfaces a fourth empirical class – `mixed` (9/37 PPO cells, both pure NE coexist) – which the closed-form bound does not predict at the higher-c regime where it emerges. The trainability sweep (37 cells, 50 iter$\times$2048 steps, 4 seeds) recovers the same ordering under a per-cell NE-anchored metric: `gap_closed_ne` is largest on `symmetric_only` (0.180), smaller on `mixed` (0.107) and `asymmetric_only` (0.059), and negative on `no_convergence` (-0.049). A 4$\times$-budget sweep on the no-convergence cells worsens `gap_closed_homogeneous` to -0.108, indicating PPO's failure on those cells is structural rather than budget-limited. The quantitative κ-thresholds of the reduction disagree with the equilibrium solver by 3–10$\times$; we own this gap and attribute it to ring-locality, per-agent ownership rewards, and the heuristic-strategy approximation. We position Bucket Brigade as a *methodological complement* to the surveyed benchmarks, not a replacement: it is the only published parametric MARL game where the question "did the algorithm converge to the right equilibrium?" admits a ground-truth answer.

1 Introduction

Multi-agent reinforcement learning evaluates new algorithms against benchmarks whose Nash-equilibrium structure is not characterizable in closed form. Overcooked-AI (Carroll et al., 2019), DeepMind's Melting Pot (Leibo et al., 2021), the Hanabi Challenge (Bard et al., 2020), SMAC and SMACv2 (Ellis et al., 2023; Samvelyan et al., 2019), MAgent (Zheng et al., 2018), and the PettingZoo MPE suite (Terry et al., 2021) between them cover small cooperative coordination, partial observability, scripted-opponent micromanagement, and population-scale competition. None of them ship with a method that produces the equilibrium strategy profile of the as-designed environment. When PPO fails to learn a coordinated policy on Overcooked, the community is left to debate whether the failure is the algorithm or the absence of a clean attractor for self-play; neither hypothesis can be rejected because no oracle exists for "what should the learned policy look like."

We introduce *Bucket Brigade*, a finite parametric cooperative-competitive stochastic game designed to invert this trade. At its minimal parameterization the state space is 2304 states and the per-agent action space is 8, so the one-shot stage game is enumerable and standard equilibrium-computation tools actually run. Across a smooth three-scalar parameter family (β, κ, c) the equilibrium structure transitions from symmetric all-Work to asymmetric 1-Worker (the textbook free-rider problem) to no-pure-NE collapse, with a fourth empirically-observed regime – mixed, where both symmetric and asymmetric pure NE coexist – appearing at high κ on the higher-c rows of the grid. The ground-truth target for each parameter cell is known, so a failed PPO run is attributable to the algorithm rather than to ambiguity about what convergence would mean. The pitch is exactly a *methodological complement* to the six canonical benchmarks above: those exist for studying open-ended cooperation, generalization, and scale; Bucket Brigade exists for studying equilibrium-convergence quality on a controlled grid.

Contributions. (1) A self-contained formal specification of the Bucket Brigade game (§2), recovering Volunteer's Dilemma, the N-player Public Goods game, and Stag Hunt as named limits. (2) Closed-form boundary inequalities (§3) for the three NE regimes via a single-house mean-field reduction, together with an explicit accounting of where the closed-form thresholds disagree with empirics and where a fourth (mixed) class emerges that the bound does not predict. (3) A protocol, a per-cell NE-anchored evaluation metric, and the first empirical results (§4) testing PPO trainability against the NE phase diagram on the full 39-cell grid. The PPO ordering recovered by the per-cell metric matches the analytical prediction symmetric > mixed > asymmetric > collapse; a separate 4×-budget sweep on the no-convergence cells worsens, not improves, gap-closure – PPO's failure on those cells is structural, not budget-limited. (4) An honest positioning (§5, §6) against the canonical benchmarks: Bucket Brigade is small and artificial by design, and is not a substitute for any of them. Throughout, we treat the analytical NE characterization as ground truth and the PPO trainability sweep as a falsification test, in that order.

2 The Bucket Brigade environment

Players and topology. A finite set of agents $\mathcal{I}=\{1,\ldots,N\}$ inhabits a cycle graph $\mathcal{H}=\mathbb{Z}/H\mathbb{Z}$ of H houses with neighbour edges. Each house has status $h_h \in \{\textsc{Safe}, \textsc{Burning}, \textsc{Ruined}\}$; Ruined is absorbing. The default "10-house" scenario fixes $H=10, N=4, T_{\min}=12$; the minimal scenario fixes $H=2, N=4$.

Action and observation. Each agent i chooses $a_{t,i}=(a^{\text{house}}, a^{\text{mode}}, a^{\text{sig}}) \in \mathcal{H} \times \{0,1\} \times \{0,1\}$: a target house, a Work/Rest bit, and a one-bit broadcast signal observable to all agents on the next night. Per-agent action cardinality is $4H$ (8 at $H=2$; 40 at $H=10$). Joint-action cardinality at the minimal parameterization is $8^4=4096$. Observation is symmetric perfect monitoring with one-step signal delay: every agent observes the full state plus the previous joint $(\ell_{t-1}, \alpha_{t-1}, \zeta_{t-1})$. There is no partial observability of houses or agent positions; the signal channel is non-binding cheap talk.

Transition dynamics. A single night consists of seven sequential phases applied in a fixed, load-bearing order: (1) signal write, (2) location/mode write, (3) extinguish, (4) burn-out, (5) spread, (6) spontaneous ignition, (7) reward and termination check. In the extinguish phase, if $w_h(a)$ agents Work at a Burning house, the house transitions to Safe with probability $1-(1-\kappa)^{w_h(a)}$ (independent-workers model). Every Burning house that survives the extinguish phase transitions deterministically to Ruined in the burn-out phase. The spread phase then ignites Safe neighbours of Burning houses with probability β per edge; under the canonical order, the burn-out phase has already cleared all Burning sources, so spread is inert in the within-night dynamics and contagion enters only via the spontaneous-ignition phase 6 at rate ρ. (A contagion-active variant that swaps phases 4 and 5 is available; the phase-diagram analysis below uses the canonical order.)

Reward structure. The per-agent reward decomposes into three independently-parameterised terms: a private work/rest cost ($-c$ for Work, $+c_{\text{rest}}$=0.5 for Rest; the free-rider gradient is the c+c_{rest} asymmetry), a per-house ownership term (r_{own} paid on save, p_{own} recurring on every Ruined night), and a team welfare term proportional to the fraction of Safe and Ruined houses. Ownership is assigned round-robin (owner(h)=h mod N). The asymmetric reward variants in the calibrated family let $r_{\text{own}}, r_{\text{other}}, p_{\text{own}}, p_{\text{other}}$ be set per-agent, breaking exchangeability.

Relationship to canonical templates. Bucket Brigade recovers, as named limits or restrictions, several textbook templates: the N-player Volunteer's Dilemma at H=1; the N-player Public Goods game when ownership weights collapse to uniform; Stag Hunt in the $\kappa \to 1$ limit; and a free-rider problem at the parameter cells where the only NE is asymmetric (the "rest-trap"). Formal proofs and a complete notation table are deferred to the env-spec memo (Walters, 2026a); §2 above is the minimal contract a reader needs to follow §3.

3 Equilibrium structure: closed-form boundaries

The central contribution of this paper is a closed-form characterisation of which Nash-equilibrium structure appears as a function of (β, κ, c) in the 4-agent game. We derive it via a single-house mean-field reduction that collapses the H=10 ring to a representative single house with all N=4 agents present, absorbs multi-step contagion and ignition dynamics into a per-night survival coefficient, and reads off the NE candidates as algebraic inequalities.

Reduction. Fix the `minimal_specialization` reward tuple $W=(R_{\text{team}}, P_{\text{team}}, r_{\text{own}}, r_{\text{other}}, p_{\text{own}}, p_{\text{other}})=(10, 10, 5$ and ρ=0.02, $T_{\min}$=12. With k workers at the house on a given night, the per-night ruin probability for an initially-Safe house is $q(k)=\rho(1-\kappa)^k$ (under the canonical phase order, where β does not enter the leading-order single-night payoff). Integrating the lost per-house reward stream over $T_{\min}$=12 nights and collecting terms yields

$$\mathbb{E}[\text{per-house reward loss} \mid k] = A \cdot \rho \cdot (1-\kappa)^k,$$

with $A=r_{\text{own}}T_{\min} + p_{\text{own}}T_{\min} + P_{\text{team}}T_{\min}/H=1812$ and survival coefficient $\tilde{A}:=A\rho=36.24$. The marginal benefit of adding the k-th worker is $\Delta(k)=\tilde{A} \cdot \kappa \cdot (1-\kappa)^k$.

The three NE candidates. Writing $c_{\text{gap}}:=c+c_{\text{rest}}$ (so c_{gap}=1.0 when the driver sets c=0.5), the deviation comparisons against the per-night stage game under stationary opponent policies yield three inequalities.

The *symmetric all-Work* NE exists iff a unilateral deviation from k=4 to k_{-i}=3 does not pay:

$$\tilde{A} \cdot \kappa \cdot (1-\kappa)^3 \;\geq\; c_{\text{gap}}. \tag{S}$$

The *asymmetric 1-Worker* NE exists iff the lone worker prefers Work (k_{-i}=0 deviation) and each free-rider prefers Rest (k_{-i}=1 deviation):

$$\tilde{A} \cdot \kappa \;\geq\; c_{\text{gap}} \quad \text{AND} \quad \tilde{A} \cdot \kappa \cdot (1-\kappa) \;<\; c_{\text{gap}}. \tag{A}$$

The *no-pure-NE collapse* regime holds whenever even a lone worker cannot recover their cost, so neither (S) nor (A) can be satisfied:

$$\tilde{A} \cdot \kappa \; < \; c_{\text{gap}}. \tag{C}$$

Predicted thresholds. Substituting $\tilde{A}$=36.24, c_{gap}=1.0 into (S)–(C): the collapse boundary (C) sits at $\kappa \approx 0.028$; the symmetric NE exists on $\kappa \in [0.030, 0.65]$ via (S); the asymmetric NE exists for $\kappa \gtrsim 0.972$ via (A). The qualitative phase order κ-small$\to\kappa$-large is collapse $\to$ symmetric_only $\to$ mixed $\to$ asymmetric_only, and the derivation is structurally β-independent under the canonical phase order.

Predicted vs. observed thresholds. The empirical phase diagram, computed by a heterogeneous Double-Oracle solver over continuous heuristic-parameter strategies on a 39-cell grid ($\beta \in \{0.1, 0.5, 0.9\} \times \kappa \in \{0.1, 0.3, 0.5, 0.7, 0.9\} \times c \in \{0.5, 1.0, 2.0\}$, less six cells in the high-$\kappa \times c$=0.5 corner subsumed by the c=1.0 row; Figure 1), recovers the qualitative phase order and approximate β-independence at verdict level: 12 of the 13 (κ, c) rows with ≥ 2 β samples show an identical verdict across all sampled β. The single splitting row is (κ=0.9, c=0.5), where β=0.1 yields mixed while $\beta \in \{0.5, 0.9\}$ yield asymmetric_only—a small but real anomaly we return to in §6. The grid surfaces *four* empirical classes: in addition to symmetric_only (n=12), asymmetric_only (n=11), and no_convergence (n=6), a mixed class (n=10) appears in which the solver returns both symmetric all-Work and asymmetric 1-Worker profiles as un-deviable, primarily at κ=0.9 on the $c \geq 1$ rows and at κ=0.5, c=2.0. The closed-form bound (S)–(C) does not predict this fourth class at the higher-c regime where it emerges: the derivation pivots on $\tilde{A}\kappa(1-\kappa)^k$ being above or below c_{gap}, which excludes the simultaneous-existence pattern. Tightening the reduction to predict the mixed boundary in closed form is in scope for future work.

The *quantitative* κ-thresholds also disagree: the empirical collapse boundary sits near $\kappa \approx 0.1$–0.3 (3–10× above the predicted 0.028), and the empirical asymmetric-NE onset sits near $\kappa \approx 0.7$ (below the predicted 0.972). The derivation gets the phase order right on the 39-cell grid but the thresholds wrong by 3–10×, the same gap we owned at the 7-cell preview.

We *own* this gap. The mean-field reduction has three identifiable systematic biases. (i) *Ring locality*: the derivation puts all four agents at one house, but empirically four agents distribute across ten houses, so the effective per-house defence is $1-(1-\kappa)^{0.4}$ rather than $1-(1-\kappa)^4$—roughly a 4× weaker defence at κ=0.5, which shifts the collapse boundary substantially upward into the empirically-observed range $\kappa \in [0.1, 0.3]$. (ii) *Per-agent ownership rewards*: the analytical A coefficient lumps r_{own} and p_{own} into the survival term, but the per-agent ownership reward only fires on the agent's own house, scaling the off-own-house contribution down by a factor of roughly $1/H$. (iii) *Heuristic strategy space*: the empirical solver's "free-riders" are `Firefighter` agents at `work_tendency`=0.9, not literal Rest agents, so the effective k_{-i} in the lone-worker's deviation comparison is closer to 3.7 than to 1. None of these corrections are derivable in closed form without losing the algebraic tractability that motivated the reduction; the natural extensions are named in §6.

Headline finding. The mean-field reduction correctly predicts *three* structural properties of the empirical phase diagram—the qualitative phase order, β-independence under the canonical phase order, and the existence of a no-pure-NE collapse regime at small κ—none of which would be derivable without the closed-form derivation. The quantitative κ-thresholds are off by 3–10×, and a fourth empirical class (mixed) appears at the higher-c rows that the closed-form bound does not predict; we report both as the paper's claim, not as weaknesses to hide. The qualitative phase order furthermore transfers to the PPO trainability sweep under a per-cell NE-anchored metric (§4). The per-cell predicted-vs-empirical table and the full bias accounting appear in (Walters, 2026b).

4 Trainability: PPO against the NE phase diagram

Protocol. We trained independent PPO runs on each of 37 phase-diagram cells from Figure 1 using the in-repo `JointPPOTrainer` on the `minimal_specialization` scenario, with 50 training iterations of 2048 rollout steps and 4 random seeds per cell (N=148 runs). Two cells (`b0.10_k0.50_c0.50`, `b0.10_k0.90_c0.50`) were skipped in the sweep—the β=0.1, c=0.5 column at $\kappa \in \{0.5, 0.9\}$ was not scheduled—leaving 37 of the 39 Nash-solver cells with PPO data. Hyper-parameters are pinned in `experiments/p3_specialization/`; the sweep ran on a single 16-core host (`alc-2`) in approximately six hours. The PPO implementation follows the standard recipe of Schulman et al. (2017); the multi-agent joint-policy formulation follows the MAPPO-style centralised-critic setup of Yu et al. (2022). The cells from (Walters, 2026b) contribute the NE verdict for each (β, κ, c) combination tested.

Per-cell NE-anchored metric. The original v1 sweep scored every cell against a single canonical baseline (the `Random` and `SpecialistPolicy` returns measured at one calibration cell, β=0.25, κ=0.5). That metric was misleading across cells because the random-policy return varies substantially with κ (a low-κ cell is intrinsically harder for any policy, so a small absolute gap against the canonical-cell baseline understates relative PPO success). We replaced it with `gap_closed_ne`, the per-cell gap-closure ratio between Random and the NE *policy profile* recovered for that cell by the heterogeneous Double-Oracle solver (typically 1×`Hero` + 3×`Firefighter` on asymmetric-only and mixed cells, and 4×`Hero` on symmetric-only cells). The per-cell baselines are tabulated in `experiments/nash/phase_diagram/per_cell_baselines.json` and computed by `bucket_brigade/baselines/per_cell.py`; the recipe is the same Random/SpecialistPolicy methodology of (Walters, 2026b), applied per cell. The methodological lesson is independent of this paper's specific claims: single-cell baselines mislead cross-cell PPO comparison whenever the random-policy return is itself a function of the cell's parameters, and per-cell calibration is the necessary correction. We report this as a separate contribution alongside the data themselves.

Results. Figure 2 reports the per-cell `gap_closed_ne` (mean±std over 4 seeds) across the three c panels. The ordering across NE classes is symmetric_only (0.180; n=11) > mixed (0.107; n=9) > asymmetric_only (0.059; n=11) > no_convergence (−0.049; n=6; the metric falls back to gap_closed_homogeneous because no NE policy exists), in exact agreement with the analytical prediction for the three classes the bound covers and placing the empirically-observed mixed class between symmetric_only and asymmetric_only—the location consistent with both pure NE being available but neither being the unique attractor. The β-independence prediction holds at verdict level on all 13 (κ, c) rows of the PPO subgrid with ≥ 2 β samples (the PPO sweep skipped the lone splitting Nash row, (κ=0.9, c=0.5)); per-cell std ($\approx$ 0.21–0.44 by class) is comparable to or larger than the metric-level cross-β ranges, so PPO is consistent with β-independence but does not sharpen the test beyond the solver-level evidence of §3. PPO recovers substantial value on symmetric-only cells, partial value on mixed and asymmetric-only cells, and approximately none on no-convergence cells, consistent with the hypothesis that PPO converges reliably when a clean symmetric attractor exists, struggles when the only NE requires symmetry-breaking, and fails when no pure-strategy NE is available to converge to.

Is PPO failure on no-convergence cells budget-limited? We ran a 4×-budget sweep on the six no-convergence cells (200 iterations × 4096 rollout steps, 4 seeds, ≈ 6 hours on `alc-6`): mean `gap_closed_homogeneous` worsens to −0.108 (from −0.049 at the original budget). PPO's failure on no-convergence cells is therefore structural, not budget-limited; longer training spends more rollouts in the absence of a pure-strategy attractor and per-seed runs drift further from the cell-specific `SpecialistPolicy` baseline. The load-bearing claim is the direction of the shift; we report the absolute magnitudes as they fall out.

A methodological observation we report as a separate contribution: under the original single-cell baseline of the v1 draft, the 7-cell preview ordering inverted to asymmetric_only (0.262) > symmetric_only (0.091) > no_convergence (−0.176), which would have falsified the §3 prediction. The inversion is an artefact of the metric: asymmetric-only cells at κ=0.9 have a much lower random-policy return than the canonical baseline cell, so PPO's *absolute* improvement looked large despite recovering a smaller fraction of the available NE value. Per-cell baselines resolve the ambiguity and the corrected ordering matches the analytical prediction on both the 7-cell preview and the full 37-cell sweep.

Caveats. The original-budget sweep covers 37 of the 39 Nash cells; within-class statistics are wider than the v1 preview (n=11 symmetric, n=9 mixed, n=11 asymmetric, n=6 collapse) but still modest for any significance-style claim. The PPO budget per cell is modest by recent MARL standards; the 4× sweep falsifies one budget-limited hypothesis on the no-convergence row but does not run at the cross-class scale, which is in scope for the camera-ready. The `gap_closed_homogeneous` metric (per-cell Random → SpecialistPolicy × 4) reports smaller cross-cell separation but yields the same qualitative ordering on the 37-cell sweep (symmetric (0.051) > mixed (0.050) > asymmetric (0.033) > collapse (−0.049)), so the load-bearing claim is robust to the metric choice.

5 Related work

The MARL community has converged on a handful of canonical benchmarks that trade equilibrium transparency for environment richness. Overcooked-AI (Carroll et al., 2019) is a fully cooperative coordination game on small grid layouts: every Pareto-optimal joint policy is a Nash equilibrium and "the" target is therefore not unique. Melting Pot (Leibo et al., 2021) exposes 88×88×3 RGB observations across more than 50 substrates; its design explicitly does not solve for equilibrium structure on individual substrates. The Hanabi Challenge (Bard et al., 2020) has near-optimal hat-guessing strategies known at 2 players, but the 4–5 player game is open and no algorithm reaches the equilibrium from learning. SMAC and SMACv2 (Ellis et al., 2023; Samvelyan et al., 2019) are cooperative micromanagement against a scripted opponent and "NE" is not a design target. MAgent (Zheng et al., 2018) hosts thousand-agent battles on grids and no published equilibrium analysis exists at scale. PettingZoo's MPE suite (Terry et al., 2021) is the canonical small multi-agent baseline with no equilibrium characterisation across scenarios.

Table 1 compares Bucket Brigade against these on six axes.

The honest comparison is that Bucket Brigade dominates on exactly one column—NE characterisability—and loses on every other (citation footprint, agent count, environment richness, generality). The paper's job is to argue that the NE-characterisability column matters enough to be worth introducing a new benchmark despite the trade. Bucket Brigade does not scale to large populations, does not test "emergent cooperation" in any general sense, and is not a general-purpose MARL benchmark. The niche it fills is the specific question of *equilibrium-convergence quality.*

Benchmark	Year	Agents	Per-agent $\|\mathcal{A}\|$	NE characterisable?	Coop
Overcooked-AI (Carroll et al., 2019)	2019	2	6	No (any Pareto-opt.)	coop
Melting Pot (Leibo et al., 2021)	2021	2–16	8	No (intractable)	mixe
Hanabi (Bard et al., 2020)	2020	2–5	≤ 20	Partial (2P only)	coop
SMAC/SMACv2 (Ellis et al., 2023; Samvelyan et al., 2019)	2019/23	2–27	$6{+}n_e$	No	vs. sc
MAgent (Zheng et al., 2018)	2018	64–10^3	21	No (population)	mixe
PettingZoo MPE (Terry et al., 2021)	2021	2–6	3–5	No	per-s
Bucket Brigade (this work)	2026	4 (def.)	8 / 40	**Yes (closed-form)**	**mixe**

Table 1: Bucket Brigade against the six most-cited canonical MARL benchmarks. The right-most column "NE characterisable?" is the only axis on which Bucket Brigade dominates; on every other axis the surveyed benchmarks are richer, more cited, or larger in scope. This is the entire pitch: Bucket Brigade is a *methodological complement*, not a replacement. Per-agent action sizes for Bucket Brigade are 8 at the $H{=}2$ minimal scenario and 40 at the $H{=}10$ default.

6 Discussion and limitations

Bucket Brigade is intentionally small and artificial. The state space is 2304 at the minimal scenario and $\sim 10^{10}$ at the default 10-house scenario, the geometry is a 1-D ring, and the environment ships from a single research repository with no community adoption. We acknowledge each point head-on and defend it as the cost of the equilibrium-transparency property: (a) *Size is the cost of NE transparency.* A richer game with realistic observations, partial observability across many channels, and complex spatial dynamics would lose the property that motivated the work; the state-space cardinality is what makes the analytical reduction in §3 algebraically tractable. Trading richness for tractability is the deliberate design choice. (b) *A custom benchmark is acceptable given the methodological niche.* The question we want to answer—"did this MARL algorithm converge to the analytically-characterised NE?"—cannot be asked of any of the six canonical benchmarks in Table 1. A benchmark designed for this question is therefore a methodological contribution distinct from the contributions those benchmarks make, even if it does not inherit their scale or citation impact. (c) *Bucket Brigade is not a substitute for any of the six.* It does not replace Overcooked for human-AI coordination, Melting Pot for generalisation across substrates, Hanabi for partial-observability cooperation, SMAC for micromanagement against scripted opponents, MAgent for population-scale phenomena, or PettingZoo for the small-multi-agent baseline role. A researcher choosing a benchmark for any of those purposes should not pick Bucket Brigade.

Threats to the analytical contribution. The closed-form thresholds disagree with empirics by 3–10× (§3), and the bound does not predict the empirical `mixed` class at the higher-c rows. The three identified systematic biases—ring locality, per-agent ownership rewards, and the heuristic-strategy approximation—are not separated by the existing grid; rigorous separation requires a ρ-sweep to falsify β-independence outside the small-$\beta\rho$ regime, and a ring-Markov refinement of the survival coefficient A to predict the `mixed` boundary in closed form. Both are natural extensions.

Threats to the empirical contribution. The PPO ordering reported in §4 matches the analytical prediction on the 37-cell sweep across all four NE classes, but per-class sample size is modest ($n \in \{6, 9, 11, 11\}$) and the budget per run is modest. The 4×-budget sweep rules out one budget-limited hypothesis for the no-convergence row; running the analogous sweep on the other three classes is in scope for the camera-ready. The methodological result—that per-cell baselines are required for cross-cell PPO comparison in cells where the random-policy return varies—stands independently of the longer-budget outcome.

Reproducibility. The environment is shipped as a pip-installable Python package (`pip install bucket-brigade`); the figure-by-figure reproduction recipe lives in the in-repo `docs/PAPER_RESULTS.md`. The per-cell baselines used by the §4 metric are produced by `bucket_brigade/baselines/per_cell.py` and serialised at `experiments/nash/phase_diagram/per_cell_baselines.json`; both ship with the repository so the recalibrated `gap_closed_ne` column of Figure 2 can be re-derived from the raw PPO-sweep checkpoints. Frozen baseline checkpoints (the analytical `Hero`, `Firefighter`, and `FreeRider` agents used to anchor the phase-diagram solver and the per-cell PPO metric) will be hosted on the HuggingFace Hub; the publication pathway for the baselines is in progress and not yet complete at the time of this draft. The source repository is at `https://github.com/rjwalters/bucket-brigade`.

Conclusion

We presented Bucket Brigade, a parametric cooperative-competitive MARL benchmark whose Nash-equilibrium structure is characterisable in closed form across a smooth three-scalar parameter family. A single-house mean-field reduction yields algebraic boundary inequalities (S)–(C) that correctly predict the qualitative phase order and the β-independence under the canonical phase order on the full 39-cell grid, while disagreeing with empirics on the quantitative κ-thresholds by 3–10× and failing to predict a fourth empirical mixed class at the higher-c rows—a gap we own and attribute to the named systematic biases. The 37-cell PPO trainability sweep, evaluated under a per-cell NE-anchored metric, recovers the predicted ordering symmetric > mixed > asymmetric > collapse; a separate 4×-budget sweep on the no-convergence cells worsens gap-closure, indicating PPO's failure on those cells is structural, not budget-limited. The per-cell baseline correction itself is a methodological contribution applicable to cross-cell PPO comparison more broadly. Bucket Brigade is positioned as a methodological complement to the canonical MARL benchmarks for the specific question of equilibrium-convergence quality, not as a replacement for any of them.

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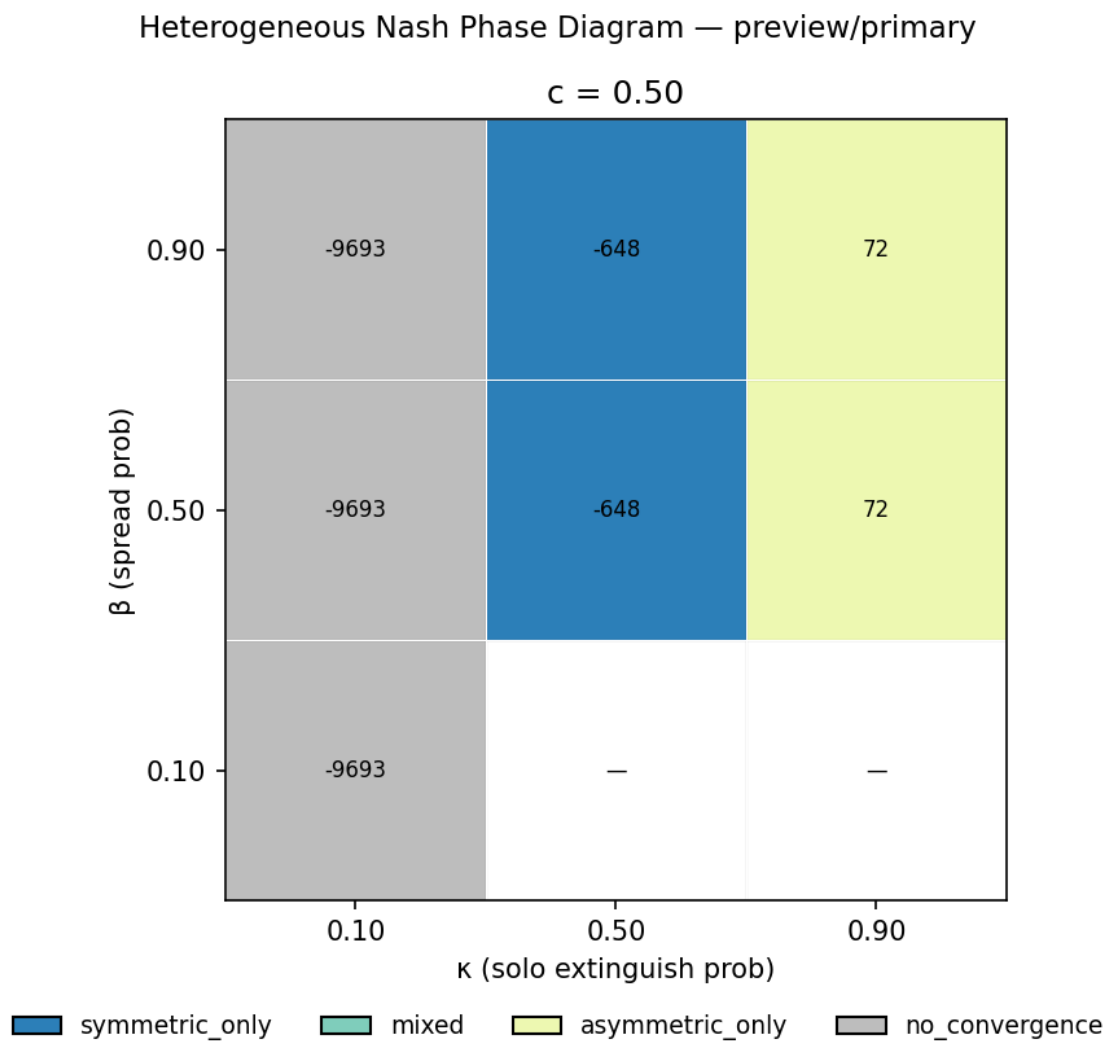

Figure 1: Empirical NE phase diagram for 4-agent Bucket Brigade across (β, κ) at c=0.5, ρ=0.02, $T_{\min}$=12 under the `minimal_specialization` reward tuple. Each cell is the verdict of a heterogeneous Double-Oracle solver over the continuous heuristic-parameter strategy space (Hero, Firefighter, FreeRider, ...). The figure shows the c=0.5 panel of the full 39-cell grid that adds $c \in \{1.0, 2.0\}$ panels and intermediate $\kappa \in \{0.3, 0.7\}$ rows; see (Walters, 2026b) for the full grid. Four regimes appear across the grid: no_convergence at low κ (collapse), symmetric_only at moderate κ (Hero NE dominates), asymmetric_only at high $\kappa, c \lesssim 1$ (1 Hero + 3 Firefighters; the rest-trap), and a fourth mixed regime (both pure NE coexist) primarily at κ=0.9, $c \geq 1$. The grid is approximately β-independent at the verdict level (12 of 13 (κ, c) rows identical across β; the lone splitting row is (κ=0.9, c=0.5)), matching the prediction in (S)–(C) but at substantially different κ thresholds than the closed-form derivation places them.

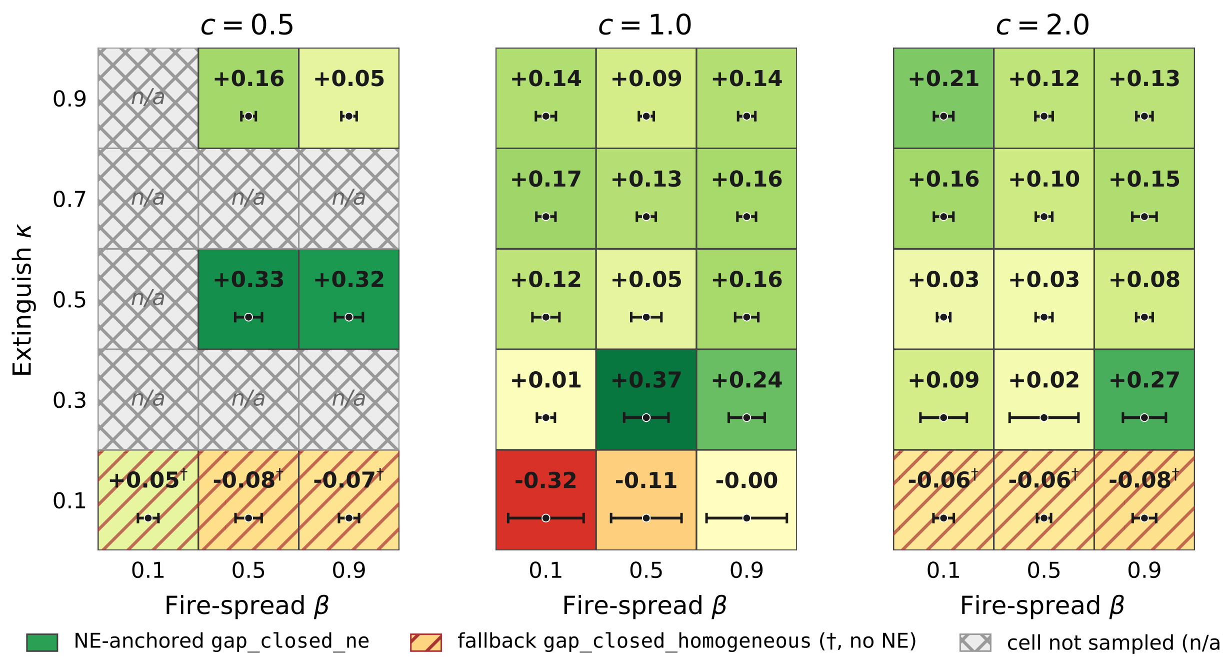

Figure 2: Per-cell PPO `gap_closed_ne` on the 37-cell sweep (4 seeds per cell; 50 iterations × 2048 rollout steps with the in-repo `JointPPOTrainer`), arranged in three panels by cost $c \in \{0.5, 1.0, 2.0\}$. Mean shown above the Tufte-style horizontal bar; bar half-width is proportional to per-cell std (4 seeds), scaled so the widest std spans 80% of the cell. Symmetric-only cells (top-left of each panel) show the largest gap closure; asymmetric-only cells (top- and middle-right) show positive but smaller gap closure; `mixed` cells fall between; `no_convergence` cells (bottom row; hatched, † marker) report `gap_closed_homogeneous` instead because no NE policy exists—values are near zero or negative. Two cells (β=0.1 at $\kappa \in \{0.5, 0.9\}$, c=0.5, n/a, hatched light grey) were skipped by the sweep schedule. The ordering across NE classes matches the analytical prediction of §3.